

04 Data Profiling and Validation

Course Name: Data Management for Machine Learning

Assignment Title: End-to-End Data Management Pipeline for a Recommendation System

Assignment: Group 51 - Data management for Machine Learning Group 51

Team Members

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1. Objective

This report documents the project's data profiling and validation stage. It captures validation code evidence, generated reports, issue logs, summaries, and the latest available data quality report from the validation folder.

2. Objective Coverage

- Apply validation checks for missing values, duplicate entries, and schema mismatch.
- Apply range and format checks where relevant to dataset rules.
- Generate a data quality report summarizing metrics, issues, and remediation evidence.
- Append the latest available data quality report PDF from reports/validation.

3. Validation Artifact Inventory

The table below lists the latest validation logs and report artifacts discovered from the project. Long file names and long paths are explicitly wrapped to avoid cut-off text in the PDF.

Artifact	Relative Path	Last Modified	Size
run_validation.txt	src\02-validation\run_validation.txt	2026-04-28 20:14:12	46.94 KB
data_quality_report_20260429T111342Z.json	reports\validation\data_quality_report_20260429T111342Z.json	2026-04-29 16:47:43	6.66 KB
data_quality_report_20260429T111342Z.pdf	reports\validation\data_quality_report_20260429T111342Z.pdf	2026-04-29 16:47:44	8.49 KB
fix_log_20260429T111342Z.csv	reports\validation\fix_log_20260429T111342Z.csv	2026-04-29 16:46:38	3.43 KB
validation_summary_initial_20260429T111342Z.csv	reports\validation\validation_summary_initial_20260429T111342Z.csv	2026-04-29 16:44:14	0.39 KB
validation_summary_revalidated_20260429T111342Z.csv	reports\validation\validation_summary_revalidated_20260429T111342Z.csv	2026-04-29 16:47:43	0.41 KB
validation_issues_initial_20260429T111342Z.csv	reports\validation\validation_issues_initial_20260429T111342Z.csv	2026-04-29 16:44:14	0.95 KB
validation_issues_revalidated_20260429T111342Z.csv	reports\validation\validation_issues_revalidated_20260429T111342Z.csv	2026-04-29 16:47:43	0.51 KB

Artifact	Relative Path	Last Modified	Size
validation_log_20260429T111342Z.jsonl	logs\validation_log_20260429T111342Z.jsonl	2026-04-29 16:47:44	0.66 KB

4. Validation Code Assets

These are the validation-related scripts, notebooks, or captured code assets discovered in the project structure.

Validation Asset	Relative Path	Last Modified	Size
run_validation_executed.ipynb	runs\prefect\20260429_115432\executed_notebooks\validate_raw\run_validation_executed.ipynb	2026-04-29 12:02:20	49.57 KB
run_validation_executed.ipynb	runs\prefect\20260429_121758\executed_notebooks\validate_raw\run_validation_executed.ipynb	2026-04-29 12:23:01	49.57 KB
run_validation_executed.ipynb	runs\prefect\20260429_132217\executed_notebooks\validate_raw\run_validation_executed.ipynb	2026-04-29 13:27:08	49.57 KB
run_validation_executed.ipynb	runs\prefect\20260429_160450\executed_notebooks\validate_raw\run_validation_executed.ipynb	2026-04-29 16:07:26	49.16 KB
run_validation_executed.ipynb	runs\prefect\20260429_160933\executed_notebooks\validate_raw\run_validation_executed.ipynb	2026-04-29 16:14:02	49.19 KB
run_validation_executed.ipynb	runs\prefect\20260429_164131\executed_notebooks\validate_raw\run_validation_executed.ipynb	2026-04-29 16:47:45	49.19 KB
run_validation.ipynb	src\02-validation\run_validation.ipynb	2026-04-29 15:06:55	46.94 KB
run_validation.txt	src\02-validation\run_validation.txt	2026-04-28 20:14:12	46.94 KB
04 Data Profiling and Validation.ipynb	src\07-submission\04 Data Profiling and Validation.ipynb	2026-04-29 20:56:59	35.04 KB

5. Validation Outcome Summary

The latest run preview indicates an initial status of **FAIL**, and a final status of **PASS**, where available.

6. Storage Structure for Validation Reports

The folder tree below documents the structure of the validation reporting area under reports/validation.

```
validation/
■■■ data_quality_report_20260428T143648Z.json
■■■ data_quality_report_20260428T144031Z.json
■■■ data_quality_report_20260428T144031Z.pdf
■■■ data_quality_report_20260429T062845Z.json
■■■ data_quality_report_20260429T062845Z.pdf
■■■ data_quality_report_20260429T065004Z.json
■■■ data_quality_report_20260429T065004Z.pdf
■■■ data_quality_report_20260429T075431Z.json
■■■ data_quality_report_20260429T075431Z.pdf
■■■ data_quality_report_20260429T085045Z.json
■■■ data_quality_report_20260429T085045Z.pdf
■■■ data_quality_report_20260429T104144Z.json
■■■ data_quality_report_20260429T104144Z.pdf
■■■ data_quality_report_20260429T111342Z.json
■■■ data_quality_report_20260429T111342Z.pdf
■■■ fix_log_20260428T144031Z.csv
■■■ fix_log_20260429T062845Z.csv
■■■ fix_log_20260429T065004Z.csv
■■■ fix_log_20260429T075431Z.csv
■■■ fix_log_20260429T085045Z.csv
```

■■■ fix_log_20260429T104144Z.csv
■■■ fix_log_20260429T111342Z.csv
■■■ validation_issues_20260428T143648Z.csv
■■■ validation_issues_initial_20260428T144031Z.csv
■■■ validation_issues_initial_20260429T062845Z.csv
■■■ validation_issues_initial_20260429T065004Z.csv
■■■ validation_issues_initial_20260429T075431Z.csv
■■■ validation_issues_initial_20260429T084831Z.csv
■■■ validation_issues_initial_20260429T085045Z.csv
■■■ validation_issues_initial_20260429T104144Z.csv
■■■ validation_issues_initial_20260429T111342Z.csv
■■■ validation_issues_revalidated_20260428T144031Z.csv
■■■ validation_issues_revalidated_20260429T062845Z.csv
■■■ validation_issues_revalidated_20260429T065004Z.csv
■■■ validation_issues_revalidated_20260429T075431Z.csv
■■■ validation_issues_revalidated_20260429T085045Z.csv
■■■ validation_issues_revalidated_20260429T104144Z.csv
■■■ validation_issues_revalidated_20260429T111342Z.csv
■■■ validation_summary_20260428T143648Z.csv
■■■ validation_summary_initial_20260428T144031Z.csv
■■■ validation_summary_initial_20260429T062845Z.csv
■■■ validation_summary_initial_20260429T065004Z.csv
■■■ validation_summary_initial_20260429T075431Z.csv
■■■ validation_summary_initial_20260429T084831Z.csv
■■■ validation_summary_initial_20260429T085045Z.csv
■■■ validation_summary_initial_20260429T104144Z.csv
■■■ validation_summary_initial_20260429T111342Z.csv
■■■ validation_summary_revalidated_20260428T144031Z.csv
■■■ validation_summary_revalidated_20260429T062845Z.csv
■■■ validation_summary_revalidated_20260429T065004Z.csv
■■■ validation_summary_revalidated_20260429T075431Z.csv
■■■ validation_summary_revalidated_20260429T085045Z.csv
■■■ validation_summary_revalidated_20260429T104144Z.csv
■■■ validation_summary_revalidated_20260429T111342Z.csv

7. Automated Validation Code Preview

The following preview captures the latest validation logic asset discovered in the project.

Source: src\02-validation\run_validation.txt

```
{
  "cells": [
    {
      "cell_type": "code",
      "execution_count": 2,
      "id": "fef32579-f890-4b53-8d24-2fcae64a4914",
      "metadata": {},
      "outputs": [
        {
          "name": "stdout",
          "output_type": "stream",
          "text": [
            "PROJECT_ROOT: C:\\Users\\barath\\recomart-pipeline\\n",
            "RAW_ROOT: C:\\Users\\barath\\recomart-pipeline\\data\\raw\\n",
            "BRONZE_ROOT: C:\\Users\\barath\\recomart-pipeline\\data\\bronze\\n",
            "\\n",
            "Discovered files:\\n",
            "- events_csv: C:\\Users\\barath\\recomart-pipeline\\data\\raw\\retailrocket\\load_date=2026-04-28\\load_hour=14\\events.csv\\n",
            "- category_tree_csv: C:\\Users\\barath\\recomart-pipeline\\data\\raw\\retailrocket\\load_date=2026-04-26\\load_hour=19\\category_tree.csv\\n",
            "- item_properties_part1_csv: C:\\Users\\barath\\recomart-pipeline\\data\\raw\\retailrocket\\load_date=2026-04-28\\load_hour=14\\item_properties_part1.csv\\n",
            "- item_properties_part2_csv: C:\\Users\\barath\\recomart-pipeline\\data\\raw\\retailrocket\\load_date=2026-04-28\\load_hour=14\\item_properties_part2.csv\\n",
            "- products_raw_json: C:\\Users\\barath\\recomart-pipeline\\data\\raw\\dummyjson\\load_date=2026-04-28\\load_hour=14\\products_raw.json\\n",
            "- categories_raw_json: C:\\Users\\barath\\recomart-pipeline\\data\\raw\\dummyjson\\load_date=2026-04-28\\load_hour=14\\categories_raw.json\\n",
            "- products_parquet: C:\\Users\\barath\\recomart-pipeline\\data\\bronze\\dummyjson\\products.parquet\\n",
            "- categories_parquet: C:\\Users\\barath\\recomart-pipeline\\data\\bronze\\dummyjson\\categories.parquet\\n",
            "\\n",
            "Initial status: FAIL\\n",
            "Final status: PASS\\n",
            "Initial issues file: C:\\Users\\barath\\recomart-pipeline\\reports\\validation\\validation_issues_initial_20260428T144031Z.csv\\n",
            "Initial summary file: C:\\Users\\barath\\recomart-pipeline\\reports\\validation\\validation_summary_initial_20260428T144031Z.csv\\n",
            "Revalidation issues file: C:\\Users\\barath\\recomart-pipeline\\reports\\validation\\validation_issues_revalidated_20260428T144031Z.csv\\n",
            "Revalidation summary file: C:\\Users\\barath\\recomart-pipeline\\reports\\validation\\validation_summary_revalidated_20260428T144031Z.csv\\n",
            "Fix log file: C:\\Users\\barath\\recomart-pipeline\\reports\\validation\\fix_log_20260428T144031Z.csv\\n",
            "JSON report file: C:\\Users\\barath\\recomart-pipeline\\reports\\validation\\data_quality_report_20260428T144031Z.json\\n",
            "PDF report file: C:\\Users\\barath\\recomart-pipeline\\reports\\validation\\data_quality_report_20260428T144031Z.pdf\\n"
          ]
        }
      ],
      "source": [
        "# DM4ML - Assignment - Validation + Auto-Fix + Revalidation + PDF Report\\n",
        "\\n",
        "import json\\n",
        "from pathlib import Path\\n",
        "from datetime import datetime, timezone\\n",
        "\\n",
        "import pandas as pd\\n",
        "\\n",
        "from reportlab.lib import colors\\n",
        "from reportlab.lib.pagesizes import A4\\n",
        "from reportlab.lib.styles import getSampleStyleSheet\\n",
        "from reportlab.platypus import SimpleDocTemplate, Paragraph, Spacer, Table, TableStyle, PageBreak\\n",
        "\\n",
        "# =====\\n",
        "# RecoMart Validation - robust file discovery + validation\\n",
        "# with remediation, revalidation, and PDF reporting\\n",
        "# =====\\n",
        "\\n",
        "PROJECT_ROOT = Path.cwd().resolve()\\n",
        "while not (PROJECT_ROOT / \"src\").exists() and PROJECT_ROOT != PROJECT_ROOT.parent:\\n",
        "    PROJECT_ROOT = PROJECT_ROOT.parent\\n",
        "\\n",
        "RAW_ROOT = PROJECT_ROOT / \"data\" / \"raw\"\\n",
        "BRONZE_ROOT = PROJECT_ROOT / \"data\" / \"bronze\"\\n",
        "\\n",
        "REMIEDIATED_RAW_ROOT = PROJECT_ROOT / \"data\" / \"remediated\" / \"raw\"\\n",
```

```

"REMEDIATED_BRONZE_ROOT = PROJECT_ROOT / \"data\" / \"remediated\" / \"bronze\"\\n\",
\"\\n\",
\"REPORT_ROOT = PROJECT_ROOT / \"reports\" / \"validation\"\\n\",
\"LOG_ROOT = PROJECT_ROOT / \"logs\"\\n\",
\"\\n\",
\"for folder in [REPORT_ROOT, LOG_ROOT, REMEDIATED_RAW_ROOT, REMEDIATED_BRONZE_ROOT]:\\n\",
\"    folder.mkdir(parents=True, exist_ok=True)\\n\",
\"\\n\",
\"UTC_NOW = datetime.now(timezone.utc)\\n\",
\"RUN_ID = UTC_NOW.strftime(\"%Y%m%dT%H%M%S\")\\n\",
\"RUN_TS = UTC_NOW.isoformat()\\n\",
\"\\n\",
\"LOG_FILE = LOG_ROOT / f\"validation_log_{RUN_ID}.json\"\\n\",
\"INITIAL_ISSUES_FILE = REPORT_ROOT / f\"validation_issues_initial_{RUN_ID}.csv\"\\n\",
\"INITIAL_SUMMARY_FILE = REPORT_ROOT / f\"validation_summary_initial_{RUN_ID}.csv\"\\n\",
\"FINAL_ISSUES_FILE = REPORT_ROOT / f\"validation_issues_revalidated_{RUN_ID}.csv\"\\n\",
\"FINAL_SUMMARY_FILE = REPORT_ROOT / f\"validation_summary_revalidated_{RUN_ID}.csv\"\\n\",
\"FIX_LOG_FILE = REPORT_ROOT / f\"fix_log_{RUN_ID}.csv\"\\n\",
\"REPORT_FILE = REPORT_ROOT / f\"data_quality_report_{RUN_ID}.json\"\\n\",
\"PDF_REPORT_FILE = REPORT_ROOT / f\"data_quality_report_{RUN_ID}.pdf\"\\n\",
\"\\n\",
\"ALLOWED_EVENT_TYPES = {\"view\", \"addtocart\", \"transaction\"}\\n\",
\"\\n\",
\"\\n\",
\"def log_event(stage, status, message, extra=None):\\n\",
\"    record = {\\n\",
\"        \"event_ts\": datetime.now(timezone.utc).isoformat(),\\n\",
\"        \"stage\": stage,\\n\",
\"        \"status\": status,\\n\",
\"        \"message\": message,\\n\",
\"        \"run_id\": RUN_ID,\\n\",
\"    }\\n\",
\"    if extra:\\n\",
\"        record.update(extra)\\n\",
\"    with open(LOG_FILE, \"a\", encoding=\"utf-8\") as f:\\n\",
\"        f.write(json.dumps(record) + \"\\n\\n\")\\n\",
\"\\n\",
\"\\n\",
\"def add_issue(issues, dataset, check_type, severity, issue_count, description,
file_path=None, phase=\"initial\"):\\n\",
\"    issues.append({\\n\",
\"        \"phase\": phase,\\n\",
\"        \"dataset\": dataset,\\n\",
\"        \"check_type\": check_type,\\n\",
\"        \"severity\": severity,\\n\",
\"        \"issue_count\": int(issue_count),\\n\",
\"        \"description\": description,\\n\",
\"        \"file_path\": str(file_path) if file_path else None,\\n\",
\"    })\\n\",
\"\\n\",
\"\\n\",
\"def add_fix(fixes, dataset, action, rows_before=None, rows_after=None, details=None,
file_path=None):\\n\",
\"    fixes.append({\\n\",
\"        \"dataset\": dataset,\\n\",
\"        \"action\": action,\\n\",
\"        \"rows_before\": rows_before,\\n\",
\"        \"rows_after\": rows_after,\\n\",
\"        \"details\": details,\\n\",
\"        \"file_path\": str(file_path) if file_path else None,\\n\",
\"    })\\n\",
\"\\n\",
\"\\n\",
\"def latest_match(base_dir: Path, pattern: str):\\n\",
\"    if not base_dir.exists():\\n\",
\"        return None\\n\",
\"    matches = list(base_dir.rglob(pattern))\\n\",
\"    if not matches:\\n\",
\"        return None\\n\",
\"    return max(matches, key=lambda p: p.stat().st_mtime)\\n\",
\"\\n\",
\"\\n\",
\"def missing_columns(df, expected_columns):

```

8. Latest run_validation Preview

Long log and code lines are manually wrapped before rendering to preserve readability.

```

{
  "cells": [
    {
      "cell_type": "code",
      "execution_count": 2,
      "id": "fef32579-f890-4b53-8d24-2fcae64a4914",
      "metadata": {},
      "outputs": [

```

```

{
  "name": "stdout",
  "output_type": "stream",
  "text": [
    "PROJECT_ROOT: C:\\Users\\barath\\recomart-pipeline\\n",
    "RAW_ROOT: C:\\Users\\barath\\recomart-pipeline\\data\\raw\\n",
    "BRONZE_ROOT: C:\\Users\\barath\\recomart-pipeline\\data\\bronze\\n",
    "\\n",
    "Discovered files:\\n",
    "- events_csv: C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\retailrocket\\load_date=2026-04-28\\load_hour=14\\events.csv\\n",
    "- category_tree_csv: C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\retailrocket\\load_date=2026-04-26\\load_hour=19\\category_tree.csv\\n",
    "- item_properties_part1_csv: C:\\Users\\barath\\recomart-pipeline\\data\\raw\\retailrock
et\\load_date=2026-04-28\\load_hour=14\\item_properties_part1.csv\\n",
    "- item_properties_part2_csv: C:\\Users\\barath\\recomart-pipeline\\data\\raw\\retailrock
et\\load_date=2026-04-28\\load_hour=14\\item_properties_part2.csv\\n",
    "- products_raw_json: C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\dummyjson\\load_date=2026-04-28\\load_hour=14\\products_raw.json\\n",
    "- categories_raw_json: C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\dummyjson\\load_date=2026-04-28\\load_hour=14\\categories_raw.json\\n",
    "- products_parquet: C:\\Users\\barath\\recomart-
pipeline\\data\\bronze\\dummyjson\\products.parquet\\n",
    "- categories_parquet: C:\\Users\\barath\\recomart-
pipeline\\data\\bronze\\dummyjson\\categories.parquet\\n",
    "\\n",
    "Initial status: FAIL\\n",
    "Final status: PASS\\n",
    "Initial issues file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_issues_initial_20260428T144031Z.csv\\n",
    "Initial summary file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_summary_initial_20260428T144031Z.csv\\n",
    "Revalidation issues file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_issues_revalidated_20260428T144031Z.csv\\n",
    "Revalidation summary file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_summary_revalidated_20260428T144031Z.csv\\n",
    "Fix log file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\fix_log_20260428T144031Z.csv\\n",
    "JSON report file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\data_quality_report_20260428T144031Z.json\\n",
    "PDF report file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\data_quality_report_20260428T144031Z.pdf\\n"
  ]
},
],
"source": [
  "# DM4ML - Assignment - Validation + Auto-Fix + Revalidation + PDF Report\\n",
  "\\n",
  "import json\\n",
  "from pathlib import Path\\n",
  "from datetime import datetime, timezone\\n",
  "\\n",
  "import pandas as pd\\n",
  "\\n",
  "from reportlab.lib import colors\\n",
  "from reportlab.lib.pagesizes import A4\\n",
  "from reportlab.lib.styles import getSampleStyleSheet\\n",
  "from reportlab.platypus import SimpleDocTemplate, Paragraph, Spacer, Table, TableStyle,
PageBreak\\n",
  "\\n",
  "# =====\\n",
  "# RecoMart Validation - robust file discovery + validation\\n",
  "# with remediation, revalidation, and PDF reporting\\n",
  "# =====\\n",
  "\\n",
  "PROJECT_ROOT = Path.cwd().resolve()\\n",
  "while not (PROJECT_ROOT / \"src\").exists() and PROJECT_ROOT != PROJECT_ROOT.parent:\\n",
  "    PROJECT_ROOT = PROJECT_ROOT.parent\\n",
  "\\n",
  "RAW_ROOT = PROJECT_ROOT / \"data\" / \"raw\"\\n",
  "BRONZE_ROOT = PROJECT_ROOT / \"data\" / \"bronze\"\\n",
  "\\n",
  "REMEDIED_RAW_ROOT = PROJECT_ROOT / \"data\" / \"remediated\" / \"raw\"\\n",
  "REMEDIED_BRONZE_ROOT = PROJECT_ROOT / \"data\" / \"remediated\" / \"bronze\"\\n",
  "\\n",
  "REPORT_ROOT = PROJECT_ROOT / \"reports\" / \"validation\"\\n",
  "LOG_ROOT = PROJECT_ROOT / \"logs\"\\n",
  "\\n",

```

9. Latest Data Quality JSON Preview

```

{
  "run_id": "20260429T111342Z",
  "run_ts": "2026-04-29T11:13:42.777701+00:00",
  "project_root": "C:\\Users\\barath\\recomart-pipeline",
  "initial_validation": {

```

```

"run_id": "20260429T111342Z",
"run_ts": "2026-04-29T11:13:42.777701+00:00",
"phase": "initial",
"project_root": "C:\\Users\\barath\\recomart-pipeline",
"raw_root": "C:\\Users\\barath\\recomart-pipeline\\data\\raw",
"bronze_root": "C:\\Users\\barath\\recomart-pipeline\\data\\bronze",
"discovered_files": {
  "events_csv": "C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\retailrocket\\load_date=2026-04-29\\load_hour=11\\events.csv",
  "category_tree_csv": "C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\retailrocket\\load_date=2026-04-29\\load_hour=11\\category_tree.csv",
  "item_properties_part1_csv": "C:\\Users\\barath\\recomart-pipeline\\data\\raw\\retailrock
et\\load_date=2026-04-29\\load_hour=11\\item_properties_part1.csv",
  "item_properties_part2_csv": "C:\\Users\\barath\\recomart-pipeline\\data\\raw\\retailrock
et\\load_date=2026-04-29\\load_hour=11\\item_properties_part2.csv",
  "products_raw_json": "C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\dummyjson\\load_date=2026-04-29\\load_hour=11\\products_raw.json",
  "categories_raw_json": "C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\dummyjson\\load_date=2026-04-29\\load_hour=11\\categories_raw.json",
  "products_parquet": "C:\\Users\\barath\\recomart-
pipeline\\data\\bronze\\dummyjson\\products.parquet",
  "categories_parquet": "C:\\Users\\barath\\recomart-
pipeline\\data\\bronze\\dummyjson\\categories.parquet"
},
"overall_status": "FAIL",
"issues_file": "C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_issues_initial_20260429T111342Z.csv",
"summary_file": "C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_summary_initial_20260429T111342Z.csv",
"datasets": [
  {
    "phase": "initial",
    "dataset": "retailrocket_events",
    "rows": 2756101,
    "columns": 5,
    "errors": 0,
    "warnings": 2734104,
    "status": "PASS"
  },
  {
    "phase": "initial",
    "dataset": "retailrocket_category_tree",
    "rows": 1669,
    "columns": 2,
    "errors": 0,
    "warnings": 25,
    "status": "PASS"
  },
  {
    "phase": "initial",
    "dataset": "retailrocket_item_properties",
    "rows": 20275902,
    "columns": 4,
    "errors": 0,
    "warnings": 0,
    "status": "PASS"
  },
  {
    "phase": "initial",
    "dataset": "dummyjson_products_raw",
    "rows": 194,
    "columns": 27,
    "errors": 0,
    "warnings": 92,
    "status": "PASS"
  },
  {
    "phase": "initial",
    "dataset": "dummyjson_categories_raw",
    "rows": 24,
    "columns": 3,
    "errors": 1,
    "warnings": 0,
    "status": "FAIL"
  },
  {
    "phase": "initial",
    "dataset": "products_parquet",
    "rows": 194,
    "columns": 32,
    "errors": 0,
    "warnings": 0,
    "status": "PASS"
  },
  {
    "phase": "initial",
    "dataset": "categories_parquet",
    "rows": 24,
    "columns": 8,

```

```

        "errors": 0,
        "warnings": 0,
        "status": "PASS"
    }
}
],
"revalidation": {
    "run_id": "20260429T111342Z",
    "run_ts": "2026-04-29T11:13:42.777701+00:00",
    "phase": "revalidated",
    "remediated_raw_root": "C:\\Users\\barath\\recomart-pipeline\\data\\remediated\\raw",
    "remediated_bronze_root": "C:\\Users\\barath\\recomart-pipeline\\data\\remediated\\bronze",
    "discovered_files": {
        "events_csv": "C:\\Users\\barath\\recomart-pipeline\\data\\remediated\\raw\\events.csv",
        "category_tree_csv": "C:\\Users\\barath\\recomart-pipeline\\data\\remediated\\raw\\category_tree.csv",
        "item_properties_part1_csv": "C:\\Users\\barath\\recomart-pipeline\\data\\remediated\\raw\\item_properties_part1.csv",
        "item_properties_part2_csv": "C:\\Users\\barath\\recomart-pipeline\\data\\remediated\\raw\\item_properties_part2.csv",
        "products_raw_json": "C:\\Users\\barath\\recomart-pipeline\\data\\remediated\\raw\\products_raw.json",
        "categories_raw_json": "C:\\Users\\barath\\recomart-pipeline\\data\\remediated\\raw\\categories_raw.json",
        "products_parquet": "C:\\Users\\barath\\recomart-pipeline\\data\\remediated\\bronze\\products.parquet",
        "categories_parquet": "C:\\Users\\barath\\recomart-pipeline\\data\\remediated\\bronze\\categories.parquet"
    },
    "overall_status": "PASS",
    "issues_file": "C:\\Users\\barath\\recomart-pipeline\\reports\\validation\\validation_issues_revalidated_20260429T111342Z.csv",
    "summary_file": "C:\\Users\\barath\\recomart-pipeline\\reports\\validation\\validation_summary_revalidated_20260429T111342Z.csv",
    "fix_log_file": "C:\\Users\\barath\\recomart-pipeline\\reports\\validation\\fix_log_20260429T111342Z.csv",
    "datasets": [
        {
            "phase": "revalidated",
            "dataset": "retailrocket_events",
            "rows": 2755641,
            "columns": 5,
            "errors": 0,
            "warnings": 2733184,
            "status": "PASS"
        },
        {
            "phase": "revalidated",
            "dataset": "retailrocket_category_tree",
            "rows": 1669,
            "columns": 2,
            "errors": 0,
            "warnings": 25,
            "status": "PASS"
        },
        {
            "phase": "revalidated",
            "dataset": "retailrocket_item_properties",
            "rows": 20275902,
            "columns": 4,
            "errors": 0,
            "warnings": 0,
            "status": "PASS"
        },
        {
            "phase": "revalidated",
            "dataset": "dummyjson_products_raw",
            "rows": 194,
            "columns": 27,
            "errors": 0,
            "warnings": 92,
            "status": "PASS"
        },
        {
            "phase": "revalidated",
            "dataset": "dummyjson_categories_raw",
            "rows": 0,
            "columns": 1,
            "errors": 0,
            "warnings": 0,
            "status": "PASS"
        },
        {
            "phase": "revalidated",
            "dataset": "products_parquet",
            "rows": 194,
            "columns": 32,
            "errors": 0,
            "warnings": 0,

```



```
    "status": "PASS"
  },
  {
    "phase": "revalidated",
    "dataset": "categories_parquet",
    "rows": 24,
    "columns": 8,
    "errors": 0,
    "warnings": 0,
    "status": "PASS"
  }
],
"pdf_report_file": "C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\data_quality_report_20260429T111342Z.pdf"
}
```

10. Initial Validation Summary Preview

phase	dataset	rows	columns	errors	warnings	status
initial	retailrocket_events	2756101	5	0	2734104	PASS
initial	retailrocket_category_tree	1669	2	0	25	PASS
initial	retailrocket_item_properties	20275902	4	0	0	PASS
initial	dummyjson_products_raw	194	27	0	92	PASS
initial	dummyjson_categories_raw	24	3	1	0	FAIL
initial	products_parquet	194	32	0	0	PASS
initial	categories_parquet	24	8	0	0	PASS

11. Revalidated Summary Preview

phase	dataset	rows	columns	errors	warnings	status
revalidated	retailrocket_events	2755641	5	0	2733184	PASS
revalidated	retailrocket_category_tree	1669	2	0	25	PASS
revalidated	retailrocket_item_properties	20275902	4	0	0	PASS
revalidated	dummyjson_products_raw	194	27	0	92	PASS
revalidated	dummyjson_categories_raw	0	1	0	0	PASS
revalidated	products_parquet	194	32	0	0	PASS
revalidated	categories_parquet	24	8	0	0	PASS

12. Initial Validation Issues Preview

phase	dataset	check_type	severity	issue_count	description	file_path
initial	retailrocket_events	missing_values	warning	2733644	Nulls in transactionid	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2026-04-29\load_hour=11\events.csv
initial	retailrocket_events	duplicates	warning	460	Duplicate rows	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2026-04-29\load_hour=11\events.csv
initial	retailrocket_category_tree	missing_values	warning	25	Nulls in parentid	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2026-04-29\load_hour=11\category_tree.csv
initial	dummyjson_products_raw	missing_values	warning	92	Nulls in brand	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2026-04-29\load_hour=11\products_raw.json
initial	dummyjson_categories_raw	schema	error	1	Missing columns: [category]	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2026-04-29\load_hour=11\categories_raw.json

13. Revalidated Issues Preview

phase	dataset	check_type	severity	issue_count	description	file_path
revalidated	retailrocket_events	missing_values	warning	2733184	Nulls in transactionid	C:\Users\barath\recomart-pipeline\data\remediated\raw\events.csv
revalidated	retailrocket_category_tree	missing_values	warning	25	Nulls in parentid	C:\Users\barath\recomart-pipeline\data\remediated\raw\category_tree.csv
revalidated	dummyjson_products_raw	missing_values	warning	92	Nulls in brand	C:\Users\barath\recomart-pipeline\data\remediated\raw\products_raw.json

14. Fix Log Preview

dataset	action	rows_before	rows_after	details	file_path
retailrocket_events	remove_invalid_event_values	2756101	2756101	Kept only allowed event types	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2026-04-29\load_hour=11\events.csv
retailrocket_events	drop_invalid_timestamps	2756101	2756101	Removed rows with invalid timestamp	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2026-04-29\load_hour=11\events.csv
retailrocket_events	drop_duplicates	2756101	2755641	Removed duplicate event rows	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2026-04-29\load_hour=11\events.csv
retailrocket_events	drop_missing_required_rows	2755641	2755641	Dropped rows missing required fields	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2026-04-29\load_hour=11\events.csv
retailrocket_category_tree	drop_missing_categoryid	1669	1669	Dropped rows with missing categoryid	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2026-04-29\load_hour=11\category_tree.csv
retailrocket_category_tree	drop_duplicates	1669	1669	Removed duplicate category rows	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2026-04-29\load_hour=11\category_tree.csv
retailrocket_item_properties	drop_missing_required_rows	20275902	20275902	Dropped rows with invalid required values	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2026-04-29\load_hour=11\item_properties_part1.csv
retailrocket_item_properties	drop_duplicates	20275902	20275902	Removed duplicate property rows	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2026-04-29\load_hour=11\item_properties_part1.csv
dummyjson_products_raw	drop_missing_required_rows	194	194	Dropped rows missing id/title/category/price	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2026-04-29\load_hour=11\products_raw.json
dummyjson_products_raw	drop_duplicate_ids	194	194	Kept first product per id	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2026-04-29\load_hour=11\products_raw.json
dummyjson_products_raw	fix_negative_prices	194	194	Clipped 0 negative prices to 0	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2026-04-29\load_hour=11\products_raw.json
dummyjson_products_raw	clip_ratings	194	194	Clipped 0 rating values into 1-5	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2026-04-29\load_hour=11\products_raw.json

dataset	action	rows_before	rows_after	details	file_path
dummyjson_products_raw	fix_negative_stock	194	194	Clipped 0 stock values to 0	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2026-04-29\load_hour=11\products_raw.json
dummyjson_categories_raw	drop_missing_category	24	0	Dropped null categories	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2026-04-29\load_hour=11\categories_raw.json
dummyjson_categories_raw	drop_blank_category	0	0	Dropped blank categories	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2026-04-29\load_hour=11\categories_raw.json

15. Validation Event Log Preview

```
{
  "event_ts": "2026-04-29T11:13:42.785703+00:00",
  "stage": "validation",
  "status": "started",
  "message": "Validation started",
  "run_id": "20260429T111342Z"
}
{
  "event_ts": "2026-04-29T11:14:14.271184+00:00",
  "stage": "remediation",
  "status": "started",
  "message": "Validation failed; starting remediation",
  "run_id": "20260429T111342Z"
}
{
  "event_ts": "2026-04-29T11:17:44.036779+00:00",
  "stage": "validation",
  "status": "completed",
  "message": "Validation workflow completed",
  "run_id": "20260429T111342Z",
  "initial_status": "FAIL",
  "final_status": "PASS",
  "pdf_report_file": "C:\\Users\\barath\\recomart-pipeline\\reports\\validation\\data_quality_report_20260429T111342Z.pdf"
}
```

16. Append Existing Data Quality Report

The latest validation PDF report found at reports\validation\data_quality_report_20260429T111342Z.pdf will be appended after this section.

17. Conclusion

This document consolidates validation code evidence, generated summaries, issue logs, fix logs, and the latest available data quality report into a single submission-ready PDF for the Data Profiling and Validation deliverable.

RecoMart Data Quality Report

Run ID: 20260429T111342Z
Generated at: 2026-04-29T11:13:42.777701+00:00
Project Root: C:\Users\barath\recomart-pipeline

1. Initial Validation

Initial overall status: **FAIL**

Initial Dataset Summary

phase	dataset	rows	columns	errors	warnings	status
initial	retailrocket_events	2756101	5	0	2734104	PASS
initial	retailrocket_category_tree	1669	2	0	25	PASS
initial	retailrocket_item_properties	20275902	4	0	0	PASS
initial	dummyjson_products_raw	194	27	0	92	PASS
initial	dummyjson_categories_raw	24	3	1	0	FAIL
initial	products_parquet	194	32	0	0	PASS
initial	categories_parquet	24	8	0	0	PASS

Initial Validation Issues

	check_type	severity	issue_count	description	file_path
	missing_values	warning	2733644	Nulls in transactionid	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2026-04-29\lo
	duplicates	warning	460	Duplicate rows	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2026-04-29\lo
ee	missing_values	warning	25	Nulls in parentid	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2026-04-29\lo
aw	missing_values	warning	92	Nulls in brand	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2026-04-29\lo
_raw	schema	error	1	Missing columns: ['category']	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2026-04-29\lo

2. Data Fixing Actions

The following remediation steps were applied automatically to failed or warning-prone datasets.

Fix Log

	rows_before	rows_after	details	file_path
event_values	2756101	2756101	Kept only allowed event types	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2020
estamps	2756101	2756101	Removed rows with invalid timestamp	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2020
	2756101	2755641	Removed duplicate event rows	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2020
quired_rows	2755641	2755641	Dropped rows missing required fields	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2020
categoryid	1669	1669	Dropped rows with missing categoryid	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2020
	1669	1669	Removed duplicate category rows	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2020
quired_rows	20275902	20275902	Dropped rows with invalid required values	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2020
	20275902	20275902	Removed duplicate property rows	C:\Users\barath\recomart-pipeline\data\raw\retailrocket\load_date=2020
quired_rows	194	194	Dropped rows missing id/title/category/price	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2020
ds	194	194	Kept first product per id	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2020
es	194	194	Clipped 0 negative prices to 0	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2020
	194	194	Clipped 0 rating values into 1-5	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2020
ck	194	194	Clipped 0 stock values to 0	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2020
ategory	24	0	Dropped null categories	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2020
gory	0	0	Dropped blank categories	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2020
	0	0	Removed duplicate categories	C:\Users\barath\recomart-pipeline\data\raw\dummyjson\load_date=2020
onze_metadata	194	194	Ensured bronze metadata columns exist	C:\Users\barath\recomart-pipeline\data\bronze\dummyjson\products.pa
onze_metadata	24	24	Ensured bronze metadata columns exist	C:\Users\barath\recomart-pipeline\data\bronze\dummyjson\categories.p

3. Revalidation

Revalidation overall status: **PASS**

Revalidated Dataset Summary

phase	dataset	rows	columns	errors	warnings	status
revalidated	retailrocket_events	2755641	5	0	2733184	PASS
revalidated	retailrocket_category_tree	1669	2	0	25	PASS
revalidated	retailrocket_item_properties	20275902	4	0	0	PASS
revalidated	dummyjson_products_raw	194	27	0	92	PASS
revalidated	dummyjson_categories_raw	0	1	0	0	PASS
revalidated	products_parquet	194	32	0	0	PASS
revalidated	categories_parquet	24	8	0	0	PASS

Revalidation Issues

dataset	check_type	severity	issue_count	description	file_path
retailrocket_events	missing_values	warning	2733184	Nulls in transactionid	C:\Users\barath\recomart-pipeline\data\remediated\raw\events.
retailrocket_category_tree	missing_values	warning	25	Nulls in parentid	C:\Users\barath\recomart-pipeline\data\remediated\raw\category.
dummyjson_products_raw	missing_values	warning	92	Nulls in brand	C:\Users\barath\recomart-pipeline\data\remediated\raw\product.

4. Before vs After

Validation Comparison

metric	before	after
overall_status	FAIL	PASS
error_count	1	0
warning_count	2734221	2733301

5. Conclusion

The initial validation run returned **FAIL**. After applying automated remediation steps and rerunning validation on the remediated files, the final status returned **PASS**. The detailed issue logs, fix log, and dataset summaries have been saved alongside this PDF.